

Noise-Robust Modification Method for Gaussian-Based Models With Application to Radar HRRP Recognition

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Abstract—In this letter, we introduce a novel noise-robust modification method for Gaussian-based models to enhance the performance of radar high-resolution range profile (HRRP) recognition under the test condition of low signal-to-noise ratio (SNR), and we develop an efficient scheme for its computation. This noise-robust modification method is implemented by revising the trained Gaussian-based model according to the estimated SNR of test HRRP. We apply the proposed method to adaptive Gaussian classifier and truncated stick-breaking hidden Markov model. Experimental results demonstrate that the proposed method can significantly improve the average recognition rate for noisy HRRP test samples while offering recognition performance comparable to that of original model for clean HRRP test samples. Moreover, even when the SNR of test HRRP samples is not precisely estimated, we can still obtain an acceptable result with the proposed method.

Index Terms—High-resolution range profile (HRRP), noise-robust recognition, radar automatic target recognition, statistical model.

I. INTRODUCTION

MANY studies [1]–[6] have focused on exploiting radar high-resolution range profile (HRRP) for target recognition in recent years. The HRRP, a real vector, is the amplitude of the coherent summations of the complex time return from target scatterers in each range cell as shown in Fig. 1. It contains abundant target information, such as target structure, scatterer distribution, etc. Statistical recognition [1]–[6] is an effective way to deal with the radar HRRP recognition problems. The procedure of statistical recognition contains the training phase and the classification phase. In the training phase, a statistical model is employed to describe the statistical characteristics of training HRRP samples from each target, and in the classification phase, the test HRRP sample is assigned to the class membership which has the maximum posterior probability.

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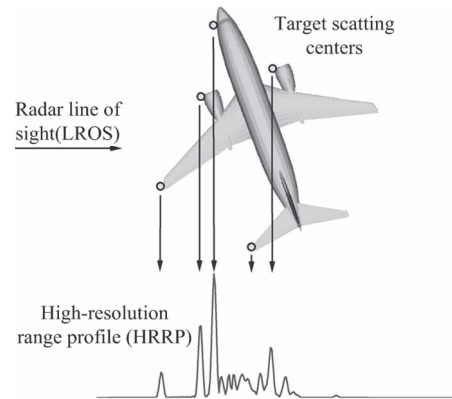


Fig. 1. Illustration of an HRRP sample from a plane target, where the scatterers on the plane target are represented in circles. This figure is cited from [7].

In practical applications, training HRRP samples usually have high signal-to-noise ratio (SNR) since they are obtained via some cooperative measurement experiments or directly via electromagnetic simulation. However, the test samples are usually got in the noncooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large distance between the noncooperative targets and radar. Previous attempts [2], [4] evaluate the effect of noise on some statistical models but do not carry out any noise-robust algorithms to deal with SNR mismatch between the trained model and noisy test HRRP samples. Since the noise component can change the statistical distribution of the noisy test HRRP sample, the recognition performance degrades dramatically for the test HRRP sample collected under low SNR and discriminated with the statistical model trained under high SNR. Therefore, developing noise-robust recognition methods is a challenging task in radar HRRP recognition field.

This letter presents a novel noise-robust modification method for Gaussian-based statistical models and applies the proposed method to adaptive Gaussian classifier (AGC) and truncated stick-breaking hidden Markov model (TSB-HMM). The experimental results show a significant gain of average recognition rate (ARR) when implementing the noise-robust modification method under the test condition of low SNR in both AGC and TSB-HMM. That is, our noise-robust modification method makes both AGC and TSB-HMM possible to recognize noisy HRRP samples. Since the SNR of target echo directly relates to the distance between the target and radar for a given noise power, our method can significantly increase the recognition distance between the target and radar in the real application.

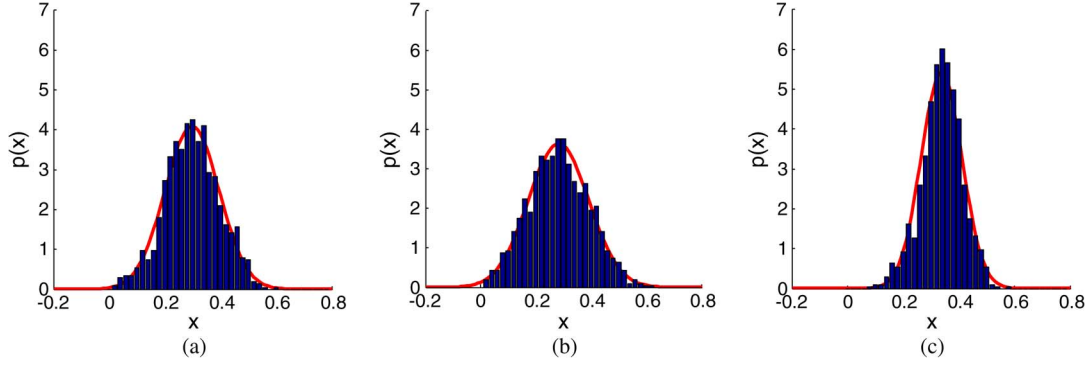


Fig. 2. PDF of echoes in some typical range cells of three targets. (a) An-26. (b) Cessna. (c) Yark-42. The blue bars represent the pdf of the echoes from the range cell approximated by the histogram method, and the red line represents the pdf estimated by the Gaussian distribution.

II. NOISE-ROBUST MODIFICATION ALGORITHM

An HRRP sample can be represented as: $\mathbf{x} = [|x_1|, \dots, |x_L|, \dots, |x_L|]$, where $|x_l|$ denotes the amplitude of the target signal in the l th range cell and L is the number of range cells in an HRRP sample.

The amplitude of the range cell $|x_l|$ is non-Gaussian distributed in theory. According to the fluctuating target model by Swerling [8], $|x_l|$ is Rayleigh distributed or Ricean distributed under the ideal condition. The ideal condition requires a large number of the scatterers with almost the same intensities per range cell, which ensures that the central limit theorem holds. Unfortunately, increasing the resolution implies a reduction of the number of scatterers per range cell; thus, the Rayleigh distribution and Ricean distribution tend to fail for the echoes under the high-resolution condition. To describe the HRRP echoes' statistical characteristics more accurately, study by Du *et al.* [4] introduces the multimodal distribution into HRRP-based statistical recognition, and study by Shi *et al.* [9] models the non-Gaussian-distributed radar HRRP data by local factor analysis. Nevertheless, these models require significant computation resources.

As an engineering compromise of describing $|x_l|$, Gaussian distribution has been widely adopted in radar HRRP statistical recognition [1], [2], [5], [6], [10], such as Gaussian-based parametric models [1], Bayesian dynamic model [2], AGC [5], hypersphere model [6], and constrained quadratic classifier [10]. Fig. 2 shows the approximated probability density functions (pdfs) of echoes in some typical range cells of three targets. From Fig. 2, we can find that the probability distribution approximated by histogram method is close to that estimated by Gaussian distribution. The advantage of adopting Gaussian distribution is that its parameter estimation is simple and its computation burden is small. The experimental results of these models [1], [2], [5], [6], [9] show that the Gaussian distribution assumption can achieve good recognition performance.

From the aforementioned discussion, the HRRP samples can be efficiently described by the Gaussian-based statistical models. The statistical recognition procedure contains two phases: the training phase and the classification phase. In the training phase, we divide the HRRP samples into frames as in [1] to avoid target-aspect sensitivity [1], [2] and train a Gaussian-based statistical model for each frame. For the training data from C targets, we train M_c independent models for each

target $c (c \in \{1, \dots, C\})$, where M_c denotes the number of frames in target c . In the classification phase, the test HRRP sample $\mathbf{x}$ will be assigned to the class with maximum posterior probability $p(c|\mathbf{x})$, where $c \in \{1, \dots, C\}$ denotes the class membership. According to Bayes' theorem, $p(c|\mathbf{x}) \propto p(\mathbf{x}|c)p(c)$, where $p(\mathbf{x}|c)$ is the class-conditional likelihood and $p(c)$ is the prior class probability. Since the prior class probability is usually assumed to be uniformly distributed [2], estimation of the posterior probability $p(c|\mathbf{x})$ of each class is turned into estimation of the class-conditional likelihood $p(\mathbf{x}|c)$ of each class. We first calculate the class-conditional likelihood $p(\mathbf{x}|c) = \max_m p(\mathbf{x}|c, m)$ for each target c , where $m = 1, \dots, M_c$ denotes the frame index for target c . Then, we assign the test HRRP sample to the class k with the maximum class-conditional likelihood $k = \arg \max_c p(\mathbf{x}|c)$, where $c = 1, \dots, C$ with C denoting the number of targets. Since the SNR of the training data is high, the effect of noise component can be neglected during the training phase. However, in the classification phase, the statistical distribution of the test HRRP sample changes because of the addition of strong noise component. The distribution of test observation cannot be approximated by Gaussian distribution under the condition of low SNR.

Our algorithm modifies the Gaussian-based models to match the noisy test HRRP sample. According to [2], for aircraft-like targets under a lookup scenario, the noises in the inphase and quadrature echoes of targets follow a Gaussian distribution. The test echo from the l th range cell under low SNR $|x_l^{\text{low}}|$ can be expressed as

$$\begin{aligned} |x_l^{\text{low}}| &= |x_l + n|, \quad |x_l| \sim N(\mu_l, \sigma_l^2) \\ \text{Re}(n) &\sim N(0, \sigma_n^2) \quad \text{Im}(n) \sim N(0, \sigma_n^2) \end{aligned} \quad (1)$$

where x_l is the complex target echo from the l th range cell, n denotes the complex noise, $N(\mu_l, \sigma_l^2)$ represents the Gaussian probability density with mean μ_l and variance σ_l^2 , and $\text{Re}(n)$ and $\text{Im}(n)$ represent the real part and the imaginary part of the complex noise n , respectively. The $|x_l^{\text{low}}|$ can be calculated via the cosine theorem

$$|x_l^{\text{low}}| = \sqrt{|x_l|^2 + |n|^2 - 2|x_l||n|\cos\varphi} \quad (2)$$

where φ denotes the angle between the complex signal x_l and the complex noise n . Since the signal x_l and the noise n are

independent, the pdf of angle φ is $f(\varphi) = 1/\pi$, $0 \leq \varphi \leq \pi$. From (1), the amplitude of noise $|n|$ is Rayleigh distributed

$$f(|n|) = \frac{|n|}{\sigma_n^2} \exp\left(-\frac{|n|^2}{\sigma_n^2}\right). \quad (3)$$

Thus, we can derive the pdf of $|x_l^{\text{low}}|$ as follows:

$$\begin{aligned} f(|x_l^{\text{low}}|) &= \int_0^\infty \int_0^\infty \frac{|x_l^{\text{low}}|}{wv} f_1(|x_l^{\text{low}}|, w, v) \\ &\quad \times f_2(w) f_3(v) dw dv \\ f_1(|x_l^{\text{low}}|, w, v) &= \frac{1}{\pi \sqrt{1 - \left(\frac{w^2 + v^2 - |x_l^{\text{low}}|^2}{2wv}\right)^2}} \\ f_2(w) &= \frac{1}{\sqrt{2\pi}\sigma_l} \exp\left(-\left(\frac{(w - \mu_l)^2}{2\sigma_l^2}\right)\right) \\ f_3(v) &= \frac{v}{\sigma_n^2} \exp\left(-\left(\frac{v^2}{2\sigma_n^2}\right)\right). \end{aligned} \quad (4)$$

However, this expression is too complicated to calculate, which means that a direct noise-robust statistical model is unfeasible to be built via (4). To solve the intractable integration in (4), we use $\sqrt{E_\varphi(|x_l^{\text{low}}|^2)}$ to approximate $|x_l^{\text{low}}|$, where $E_\varphi(|x_l^{\text{low}}|^2)$ denotes the expectation value of $|x_l^{\text{low}}|^2$ over the distribution on φ

$$\begin{aligned} |x_l^{\text{low}}| &\approx \sqrt{E_\varphi(|x_l^{\text{low}}|^2)} = \sqrt{\int_0^\pi f(\varphi) |x_l^{\text{low}}|^2 d\varphi} \\ &= \sqrt{|x_l|^2 + |n|^2}. \end{aligned} \quad (5)$$

By combining (1), (3), and (5), we can derive the pdf of $|x_t^{\text{low}}|$ as follows:

$$\begin{aligned} f(|x_t^{\text{low}}|) &= \frac{|x_t^{\text{low}}|}{\sqrt{2\pi}\sigma_l\sigma_n^2} \int_0^\pi \frac{|x_l^{\text{low}}|}{\sqrt{2\pi}\sigma_l\sigma_n^2} \\ &\quad \times \exp\left(-\left(\frac{(u - \mu_l)^2}{2\sigma_l^2} + \frac{|x_l^{\text{low}}|^2 - u^2}{2\sigma_n^2}\right)\right) du. \end{aligned} \quad (6)$$

As we can see, (6) is much simpler than (4), and hence, it provides a computationally tractable way to build a noise-robust statistical model. Since (6) reflects the effect of noise on each range cell of the HRRP, we can substitute (6) for the original Gaussian-distributed model to match the distribution of noisy test HRRP samples.

III. TWO APPLICATIONS OF NOISE-ROBUST MODIFICATION METHOD

Here, we consider two common Gaussian-based models, namely, AGC [5] and TSB-HMM [11], [12], in radar HRRP statistical recognition field as examples to demonstrate the implementation of our noise-robust modification method.

A. AGC

The AGC is a system developed by Hughes Aircraft Corporation for radar HRRP recognition [5]. In AGC, each range

cell of HRRP sample $|x_l|$ is considered to be independent Gaussian distributed with the parameters $\{\mu_l, \sigma_l\}$, where μ_l and σ_l represent the mean and the standard deviation of the Gaussian pdf for the l th range cell. In the training phase, we build independent AGC models for each frame from all targets and estimate $\theta^{(c,m)} = \{\mu_l^{(c,m)}, \sigma_l^{(c,m)}\}_{l=1}^L$ for frame m of target c ($c = 1, \dots, C$ and $m = 1, \dots, M_c$ where C denotes the number of targets and M_c denotes the number of frames in target c). In the classification phase, given frame m of target c , the likelihood of a clean test HRRP sample using original AGC can be expressed as

$$p(\mathbf{x}|c, m) = \prod_{l=1}^L \frac{1}{\sqrt{2\pi}\sigma_l^{(c,m)}} \exp\left(-\frac{(|x_l| - \mu_l^{(c,m)})^2}{2\sigma_l^{(c,m)^2}}\right). \quad (7)$$

As we discussed in Section II, the distribution of $|x_l|$ cannot be approximated by Gaussian distribution under the condition of low SNR. According to (6), given frame m of target c , the likelihood of a noisy test HRRP sample using noise-robust AGC can be expressed as

$$\begin{aligned} p(\mathbf{x}|c, m) &= \prod_{l=1}^L \frac{|x_l^{\text{low}}|}{\sqrt{2\pi}\sigma_l^{(c,m)}\sigma_n^2} \int_0^\pi \frac{|x_l^{\text{low}}|}{\sqrt{2\pi}\sigma_l^{(c,m)}\sigma_n^2} \\ &\quad \times \exp\left(-\left(\frac{(u - \mu_l^{(c,m)})^2}{2\sigma_l^{(c,m)^2}} + \frac{|x_l^{\text{low}}|^2 - u^2}{2\sigma_n^2}\right)\right) du. \end{aligned} \quad (8)$$

Then, the test HRRP sample will be assigned to the class k with the maximum class-conditional likelihood as we mentioned in Section II.

B. TSB-HMM

We consider each HRRP sample $\mathbf{x} = [|x_1|, \dots, |x_l|, \dots, |x_L|]$ as a 1-D observation sequence in TSB-HMM [11], [12]. The TSB-HMM for HRRP data can be expressed as $\Phi = \{\mathbf{w}_0, \mathbf{W}, \theta\}$, where $\mathbf{w}_0 = \{\omega_{0,i}\}$ is the initial state probability mass function (PMF) with $\omega_{0,i}$ denoting the initial state probability of state i , $\mathbf{W} = \{\omega_{i,j}\}$ is state transition matrix with $\omega_{i,j}$ denoting the transition probability of state i to state j , and s_l is hidden state at range cell l . In TSB-HMM, each row of the state transition matrix $\{\mathbf{w}_i\}_{i=1}^I$ and the initial state PMF $\mathbf{w}_0$ are constructed by the finite stick-breaking prior that can induce a sparse utilization of the underlying states. $\theta = \{\mu_i, \sigma_i\}_{i=1}^I$ with μ_i and σ_i denoting the mean and the standard deviation of the Gaussian pdf corresponding to state i . The TSB-HMM uses a Gaussian-distributed observation model to describe the observation $|x_l|$. In the training phase, we estimate the $\Phi^{(c,m)} = \{\mathbf{w}_0^{(c,m)}, \mathbf{W}^{(c,m)}, \theta^{(c,m)}\}$ for frame m of target c ($c = 1, \dots, C$ and $m = 1, \dots, M_c$, with C denoting the number of targets and M_c denoting the number of frames in target c).

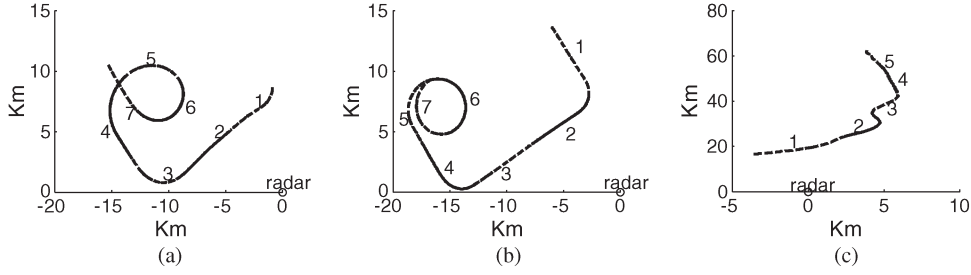


Fig. 3. Projections of target trajectories onto the ground plane. (a) An-26. (b) Cessna. (c) Yark-42.

In the classification phase, given frame m of target c , the likelihood of a clean test HRRP sample using original TSB-HMM can be expressed as

$$\begin{aligned}
 p(\mathbf{x}|c, m) &= \sum_{\mathbf{s}} \left\langle \omega_{0, s_1^{(c, m)}}^{(c, m)} \right\rangle \prod_{l=2}^L \left\langle \omega_{s_{l-1}^{(c, m)}, s_l^{(c, m)}}^{(c, m)} \right\rangle \\
 &\quad \times \prod_{l=1}^L f^{(c, m)}(|x_l| | \left\langle \theta_{s_l^{(c, m)}} \right\rangle) \\
 &\quad f^{(c, m)}(|x_l| | \theta_{s_l^{(c, m)}}) \\
 &= \frac{1}{\sqrt{2\pi} \sigma_{s_l^{(c, m)}}^{(c, m)}} \exp \left(- \frac{(|x_l| - \mu_{s_l^{(c, m)}}^{(c, m)})^2}{2\sigma_{s_l^{(c, m)}}^{(c, m)2}} \right) \quad (9)
 \end{aligned}$$

where $\langle \cdot \rangle$ means the posterior expectation for the latent variable over the corresponding distribution on it, e.g., $\langle \omega_{0, s_1^{(c, m)}}^{(c, m)} \rangle$ denotes the posterior expectations of initial state probability for the frame m of target c , $\langle \omega_{s_{l-1}^{(c, m)}, s_l^{(c, m)}}^{(c, m)} \rangle$ denotes the posterior expectations of state transition probability from state $s_{l-1}^{(c, m)}$ to state $s_l^{(c, m)}$ for the frame m of target c , and $\langle \theta_{s_l^{(c, m)}} \rangle$ denotes the posterior expectations of the observation model parameters associated with state $s_l^{(c, m)}$, with the corresponding state indicator for the l th range cell $s_l^{(c, m)} \in \{1, \dots, I\}$. The observation model $f^{(c, m)}(\cdot)$ is assumed to be Gaussian distributed for a clean test HRRP sample. Considering the variation of distribution caused by noise, given frame m of target c , the likelihood of a noisy test HRRP sample using noise-robust TSB-HMM can be expressed as

$$\begin{aligned}
 p(\mathbf{x}|c, m) &= \sum_{\mathbf{s}} \left\langle \omega_{0, s_1^{(c, m)}}^{(c, m)} \right\rangle \prod_{l=2}^L \left\langle \omega_{s_{l-1}^{(c, m)}, s_l^{(c, m)}}^{(c, m)} \right\rangle \\
 &\quad \times \prod_{l=1}^L f^{(c, m)}(|x_l^{\text{low}}| | \left\langle \theta_{s_l^{(c, m)}} \right\rangle) \\
 &\quad f^{(c, m)}(|x_l^{\text{low}}| | \theta_{s_l^{(c, m)}}) \\
 &= \frac{|x_l^{\text{low}}|}{\sqrt{2\pi} \sigma_{s_l^{(c, m)}}^{(c, m)} \sigma_n^2} \int_0^{|x_l^{\text{low}}|}
 \end{aligned}$$

$$\times \exp \left(- \left(\frac{(u - \mu_{s_l^{(c, m)}}^{(c, m)})^2}{2\sigma_{s_l^{(c, m)}}^{(c, m)2}} + \frac{|x_l^{\text{low}}|^2 - u^2}{2\sigma_n^2} \right) \right) du. \quad (10)$$

Then, the test HRRP sample will be assigned to the class k with the maximum class-conditional likelihood as we mentioned in Section II.

IV. EXPERIMENTAL RESULTS

We examine the performance of our method using the measured HRRP data of three aircrafts. The projections of target trajectories onto the ground plane are segmented and shown in Fig. 3. To test the generalization performance of the recognition methods, we select the second and fifth segments of Yark-42, the sixth and seventh segments of Cessna, and the fifth and sixth segments of An-26 as training HRRP samples and select the rest segments as test HRRP samples in our experiments. To deal with the target-aspect sensitivity, we divide the training data from Yark-42, Cessna, and An-26 into 35, 50, and 50 frames, respectively, with each frame containing 128 HRRP samples.

The SNR in the raw measured data is high, so we generate noisy test samples with different SNRs by adding Gaussian white noises to the raw test HRRP data. The SNR is defined as

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\sum_{l=1}^L P_{x_l}}{L \times \sigma_n^2} \right) \quad (11)$$

where P_{x_l} denotes the power of the original test HRRP data in the l th range cell, L is the number of range cells, and σ_n^2 denotes the power of noise. In the practical applications, we can estimate the noise power via the echoes in the range cells far away from target and then calculate the SNR of the HRRP data.

A. Noise-Robust Modification Method With True Test SNR

In this experiment, four models are constructed to evaluate the recognition performance of the noise-robust modification method, i.e., the original AGC and TSB-HMMs and the noise-robust AGC and TSB-HMMs with the true test SNR.

Fig. 4 shows the ARR versus SNRs, obtained via the four models. Since the noise-robust modification method considers the noise component of test data, it largely improves the recognition performance under low test SNR. The recognition performances of the noise-robust AGC and TSB-HMMs are significantly better than those of the original AGC and TSB-HMMs under low-SNR cases ($\text{SNR} \leq 25$ dB). Note that, when

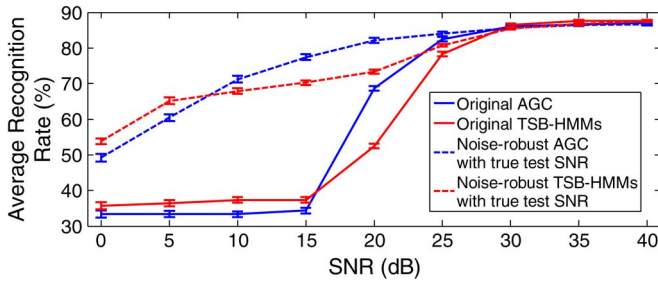


Fig. 4. Variation of the recognition performance with SNR using four models, i.e., the original AGC and TSB-HMMs and the noise-robust AGC and TSB-HMMs with the true test SNR. The ARR are averaged over ten runs, and the error bars denote the corresponding standard deviations.

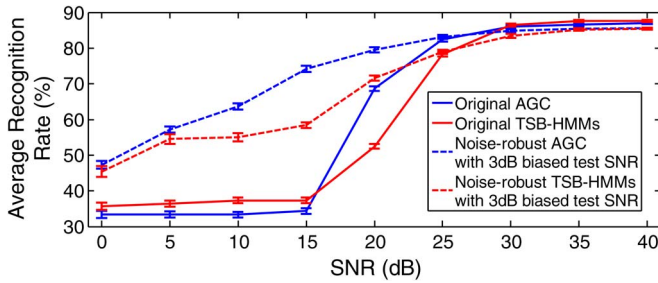


Fig. 5. Variation of the recognition performance with SNR using four models, i.e., the original AGC and TSB-HMMs and the noise-robust AGC and TSB-HMMs with 3 dB deviated from the true test SNR. The ARR are averaged over ten runs, and the error bars denote the corresponding standard deviations.

the SNR of test HRRP samples is 10 dB, original AGC and TSB-HMMs fail to recognize the target, while the ARR of noise-robust AGC and TSB-HMMs are 71.3% and 67.9%, respectively. With the increase of SNR, the influence of the noise becomes small, and the recognition performances of noise-robust AGC and TSB-HMMs are almost the same as those of the original AGC and TSB-HMMs.

B. Noise-Robust Modification Method With 3-dB-Biased Test SNR

In practical applications, it is difficult to precisely estimate the SNR of a test HRRP; thus, a bias between the estimation and the ground truth of the test HRRP's SNR should be considered. In this experiment, we consider the case of 3-dB bias between the estimation and the ground truth of the test HRRP's SNR. Here, the original AGC and TSB-HMMs and the noise-robust AGC and TSB-HMMs with 3-dB-biased test SNR are constructed to evaluate the recognition performance.

The recognition performance curves versus SNR of the four models are shown in Fig. 5. The final performances of the models are the average performance using the SNR with +3- and -3-dB deviations from the ground truth. Fig. 5 shows that the noise-robust modification method works well with the 3-dB-biased SNR. Compared with the noise-robust models with true test SNR shown in Fig. 4, the noise-robust models with 3-dB-biased SNR have a moderate decline when $\text{SNR} \leq 15$ dB. Nevertheless, the ARR of the noise-robust AGC and TSB-HMMs with 3-dB-biased SNR are also much higher than those of the original AGC and TSB-HMMs under low-SNR cases ($\text{SNR} \leq 25$ dB). Note that, when the SNR of test

HRRP samples is 10 dB, original AGC and TSB-HMMs fail to recognize the target, while the ARR of noise-robust AGC and TSB-HMMs with 3-dB-biased SNR are 63.6% and 55.1%, respectively. For high-SNR cases ($\text{SNR} \geq 30$ dB), the ARR of the noise-robust AGC and TSB-HMMs with 3-dB-biased SNR is slightly lower than that of the original AGC and TSB-HMMs because of the SNR bias.

V. CONCLUSION

We have presented a new noise-robust modification method for the Gaussian-based radar HRRP recognition model to deal with the model mismatching between clean training data and noisy test data. The proposed method can revise the trained Gaussian-based model according to the estimated SNR of test HRRP.

We further apply the proposed noise-robust modification method to two traditional HRRP statistical models, i.e., AGC and TSB-HMM. The experimental results show that the proposed method can significantly enhance the performance of radar HRRP recognition under the test condition of low SNR while providing comparable recognition performance under the condition of high SNR in both AGC and TSB-HMM. When the estimated SNR of test HRRP sample is 3 dB deviated from the real SNR of the test HRRP, our method still has acceptable recognition performance. Since the SNR of the test HRRP is related to the radar-target distance, our modification method can greatly expand the recognition range of radar.

REFERENCES

- [1] L. Du, H.-W. Liu, and Z. Bao, "Radar HRRP statistical recognition: Parametric model and model selection," *IEEE Trans. Signal Process.*, vol. 56, no. 5, pp. 1931–1944, May 2008.
- [2] L. Du, P.-H. Wang, H.-W. Liu, M. Pan, F. Chen, and Z. Bao, "Bayesian spatiotemporal multitask learning for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 59, no. 7, pp. 3182–3196, Jul. 2011.
- [3] F. Zhu, X.-D. Zhang, Y.-F. Hu, and D. Xie, "Nonstationary hidden Markov models for multispect discriminative feature extraction from radar targets," *IEEE Trans. Signal Process.*, vol. 55, no. 5, pp. 2203–2214, May 2007.
- [4] L. Du, H.-W. Liu, Z. Bao, and J.-Y. Zhang, "A two-distribution compounded statistical model for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 54, no. 6, pp. 2226–2238, Jun. 2006.
- [5] S. P. Jacobs and J. A. O. Sullivan, "Automatic target recognition using sequences of high resolution radar range-profiles," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 36, no. 2, pp. 364–381, Apr. 2000.
- [6] L. Du, H.-W. Liu, and Z. Bao, "Radar HRRP statistical recognition based on hypersphere model," *Signal Process.*, vol. 88, no. 5, pp. 1176–1190, May 2008.
- [7] J. Zwart, R. Heiden, S. Gelsema, and F. Groen, "Fast translation invariant classification of HRR range profiles in a zero phase representation," *Proc. Inst. Elect. Eng.—Radar Sonar Navig.*, vol. 150, no. 6, pp. 411–418, Feb. 2004.
- [8] P. Swerling, "Probability of detection for fluctuating targets," *IRE Trans. Inf. Theory*, vol. 6, no. 2, pp. 269–308, Apr. 1960.
- [9] L. Shi, P.-H. Wang, H.-W. Liu, L. Xu, and Z. Bao, "Radar HRRP statistical recognition with local factor analysis by automatic Bayesian Ying-Yang harmony learning," *IEEE Trans. Signal Process.*, vol. 59, no. 2, pp. 610–617, Feb. 2011.
- [10] R. A. Mitchell and J. J. Westerkamp, "Robust statistical feature based aircraft identification," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 35, no. 3, pp. 1077–1094, Jul. 1999.
- [11] J. Paisley and L. Carin, "Hidden Markov models with stick breaking priors," *IEEE Trans. Signal Process.*, vol. 57, no. 10, pp. 3905–3917, Oct. 2009.
- [12] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proc. IEEE*, vol. 77, no. 2, pp. 257–286, Feb. 1989.